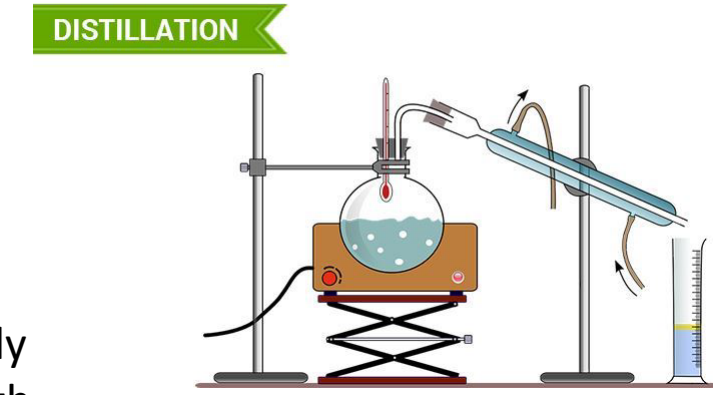
## What if I care about fast inference of uncertainty estimates?

industrial applications: embedded systems, high-load servers, etc.



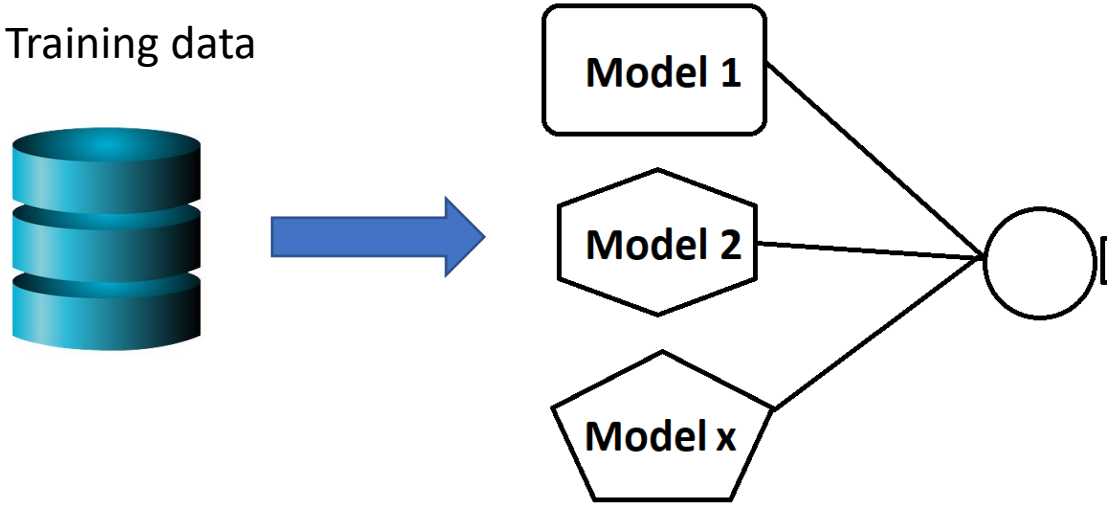
# Simple&Effective Techniques: Efficient Ensemble

## Efficient Ensemble: Distillation

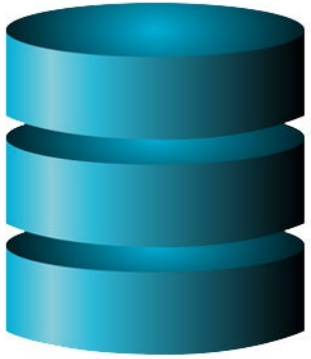



Training data

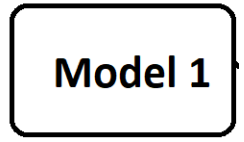
Train an ensemble




Make (a potentially larger) dataset with softlabels



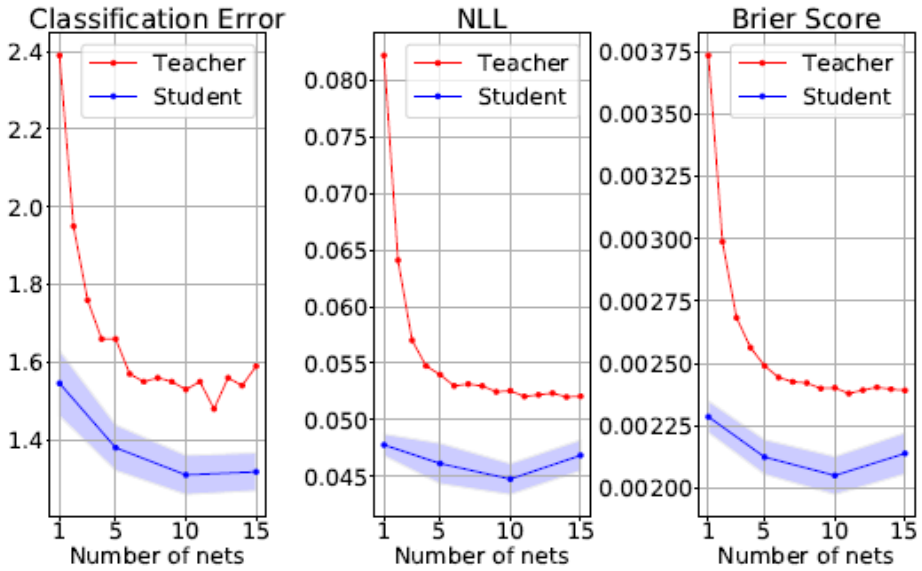
Train a student



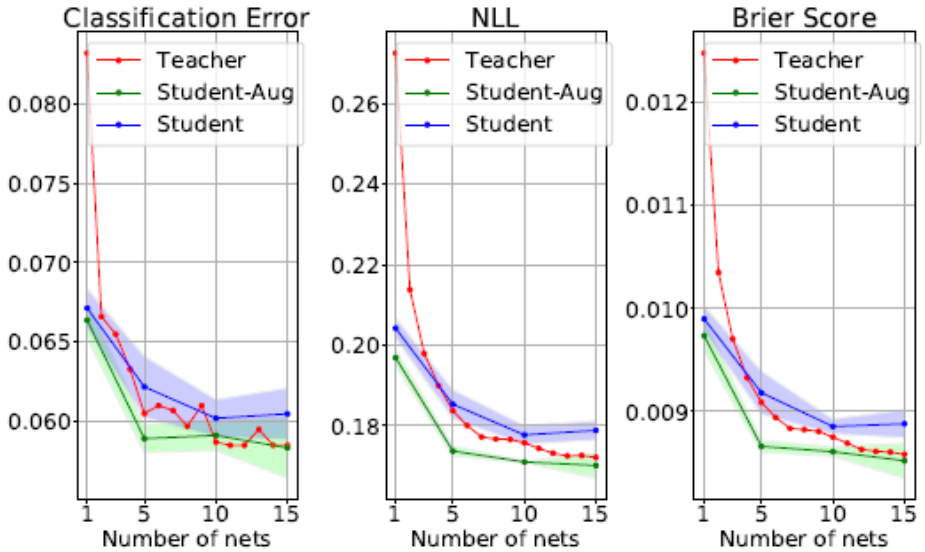

[Hinton et al. “Distilling the Knowledge in a Neural Network”, NIPS 2014]

# Simple&Effective Techniques: Efficient Ensemble

## Efficient Ensemble: Distillation




(a) MNIST




(b) CIFAR-10

[Englesson&Azizpour, "Efficient Evaluation-Time Uncertainty Estimation by Improved Distillation", ICML 2019 Uncertainty workshop]

[Furlanello et al., "Born Again Neural Networks", ICML 2018]

# Simple&Effective Techniques: Efficient Ensemble

## Efficient Ensemble: many other works on distillation

- ["Bayesian Dark Knowledge", NIPS 2015]
- ["Efficient Evaluation-Time Uncertainty Estimation by Improved Distillation", ICML UDL 2019]
- ["Ensemble Distribution Distillation", ICLR 2020]
- ["A General Framework for Ensemble Distribution Distillation", MLSP 2020]
- ["Generalized Bayesian Posterior Expectation Distillation for Deep Neural Networks" UAI 2020]
- ["Ensemble Distillation for Structured Prediction: Calibrated, Accurate, Fast-Choose Three", EMNLP 2020]
- ["Distilling Ensembles Improves Uncertainty Estimate", AABI 2021]
- ["Real-Time Uncertainty Estimation in Computer Vision via Uncertainty-Aware Distribution Distillation", WACV 2021]



# Simple&Effective Techniques: Single-Model Techniques

## Single-Model Techniques

- ["Averaging Weights Leads to Wider Optima and Better Generalization", UAI 2018]
- ["Deterministic Variational Inference for Robust Bayesian Neural Networks", ICLR 2019]
- ["Single-Model Uncertainties for Deep Learning", NeurIPS 2019]
- ["Simple and Principled Uncertainty Estimation with Deterministic Deep Learning via Distance Awareness", NeurIPS 2020]
- ["Depth Uncertainty in Neural Networks", UDL 2020]
- ["Training Independent Subnetworks for Robust Prediction", ICLR 2021]

